

Stat 310: Report on COVID-19 in New Jersey

Camila Nunez Polanco, Sarah Suratt

Intro

Background Info and Demographics

Background

New Jersey, located in the Northeastern area of the United States, consists of 21 counties.

Burlington is the largest county, and Hudson is the smallest county. New Jersey's urban counties tend to be smaller, while the more rural ones are usually larger. At one point, New Jersey had the nation's highest COVID-19 death rate per capita.

Demographics

New Jersey has a population of 9.501 million and is also the most densely populated state, with around 1,253 people per square mile. It is titled "the diner capital of the world", with around 450 diners scattered throughout. It is close to New York, so its tourism is enhanced due to its proximity. In 2022, New Jersey had about 114.6 million tourists visit, 18 million more than in 2021.

Important Events Timeline of COVID-19

March 2, 2020: The first confirmed case in New Jersey was in Bergen County. This led to a whirlwind of initiative actions from the government in New Jersey.

March 9, 2020: Governor Phil Murphy declared a state of emergency. This meant that schools shut down and either went virtual or closed completely. This helped to contain the virus from spreading to children and/or families.

March 10, 2020: The very next day, after the state of emergency was put in place in New Jersey, the first death across the state occurred from the virus.

March 16, 2020: A statewide curfew was enacted. This caused businesses (restaurants, stores, bars, casinos, etc.) to shut down. Restaurants and bars remained open only for takeout or delivery.

March 21, 2020: Governor Murphy announced a statewide stay-at-home order due to COVID-19 cases surpassing 1,000. This order made non-essential businesses close indefinitely by 9 pm that day. Social distancing, capacity restrictions, and mask mandates followed closely after this order. May 12, 2021: All of the laws enacted to combat the pandemic by Governor Murphy ultimately helped the state in containing the virus. Even with these efforts, New Jersey reached the 9th-highest number of confirmed cases and the 11th-highest number of confirmed cases per

capita in the nation. The state also reached the sixth-highest count of deaths related to the virus and the highest count per capita.

Goals

New Jersey is a smaller state, but it is densely packed, which can greatly affect the state's reaction to the COVID-19 pandemic. New Jersey is a densely populated and highly interconnected state with significant variation in urbanization, mobility patterns, and demographic composition. As such, there are likely multiple county-level factors that influenced the speed and severity of early COVID-19 spread across the state. Thus, two questions of interest in particular emerge: determining which measurable features, such as mobility, urbanization, and demographic characteristics, could help explain differences in outbreak dynamics between counties.

Additionally, given New Jersey's proximity to major metropolitan cities like New York City and Philadelphia, an important question is how these factors interacted with external influences to shape transmission patterns. By analyzing these relationships, this study seeks to identify which variables were most predictive of COVID-19 spread in New Jersey's distinct regional contexts.

Exploratory Variables

Variable	Description
Total Cases	Number of total cases
Total Deaths	Number of total deaths
Cases per 100k	Number of cases per 100k population
Deaths per 100k	Number of deaths per 100k population
Days to 100th Case	How many days it took from the first reported case to the 100th case

Figure 1: COVID-related variables

Variable	Description	Mean
Median Household Income	Represents the median income of individuals within each county	84634.5
Population Density	Measures how densely populated a county is	928.5
Unemployment Rate	Percentage of the labor force that is unemployed	7.5
Percent of Commuters	Percent of workers who commute 60+ minutes	13.2
Airport Access	Binary indicator if county contains a major airport	
Percent of Internet Access	Percent of households with Internet Access	91.8
Proximity to Major Airports	Categorizes counties based on how close they are to commercial airports	
Percent of People Age 65+	High Risk Population	23.7
Percent Uninsured	The percentage of the county's population under age 65 that does not have health insurance coverage	17.7
Percent White	Percent of White People	51.94
Percent Black	Percent of Black People	12.2
Percent Hispanic	Percent of Hispanic People	21.9
Percent Asian	Percent of Asian People	9.8
Beds per Capita	Total Staffed Beds per 1,000	2.12

Figure 2: Non-COVID-related variables to explore by county

Overall, we considered 20 variables. 7 were COVID-related, such as cases, deaths, days to 100 cases, etc., as shown in Figure 1. These non-COVID variables (Figure 2) were chosen to investigate underlying factors that may explain variations in COVID-19 outcomes across counties. Socioeconomic, demographic, and infrastructural characteristics are particularly relevant for understanding disparities in case rates, mortality, and other pandemic-related metrics. The primary focus of the study is to examine how these non-COVID variables correlate with or predict COVID-19 outcomes, shedding light on the broader social and economic determinants of pandemic severity.

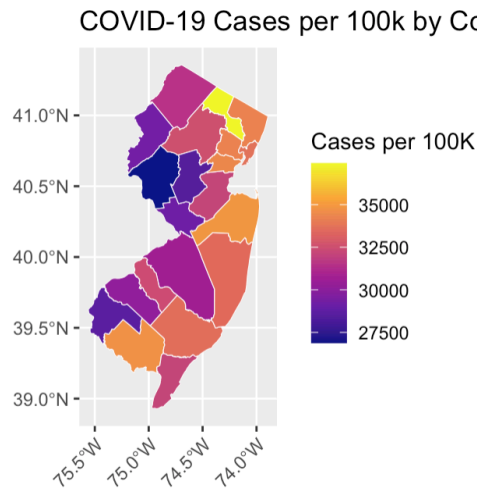
Summary Statistics

Variables	Mean	SD	p0	p25	p50	p75	p100
Final Cases	145514	94,847	18,650	52,987	142,308	216,686	327,464
Final Deaths	1719	1173	265	665	1,379	2,500	3781
Percent of Long Commuters	13.3	4.14	7.37	9.48	13.4	16.7	20.8
Median Income	84,635	25,050	45,000	72,000	80,000	90,000	133,534
Population Density	929	452	300	500	950	1200	2000
Unemployment Rate	7.6	6.38	3.9	5.1	6	7.3	34.4
Percent with Internet	91.9	2.37	87.8	90.7	91.7	94.0	95.5
Percent White	58.1	18.7	27.9	41.9	54.5	75.2	84.4
Percent Black	11.0	8.10	2.27	5.29	9.96	15.8	36.0
Percent Hispanic	19.4	10.9	0.654	10.7	17.0	23.0	43.4
Percent Asian	7.50	6.68	7.84	2.96	5.51	10.6	25.0
Percent 65 and over	17.7	3.36	12.4	15.7	17.8	18.9	28.6
Days to 100	14.6	8	8	11	14	18	26
Cases per 100k	32,128	2,632	26,880	30,188	32,443	34,244	37,429
Deaths per 100k	376	63.1	239	345	372	408	509
Percent Uninsured	23.7	7.76	14.1	16.8	21.1	27.4	38.9
Total Staffed Beds per 1,000	2.12	1.14	0.95	1.45	1.83	2.51	6.14

Figure 3: Summary Statistics

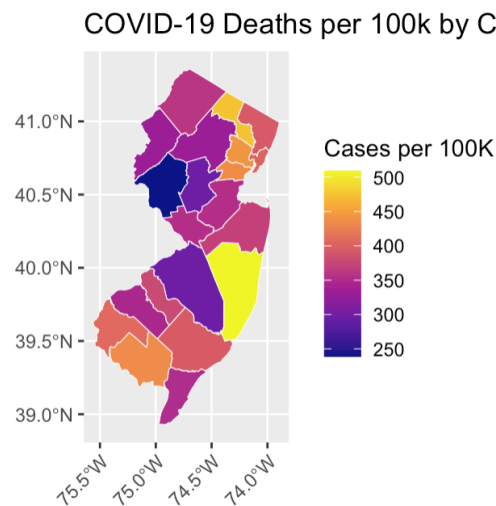
This table shows significant disparities in COVID-19 outcomes and socioeconomic factors across counties. Urban counties with higher population density (mean = 929 people/sq mi) and lower median income (\$84,635) experienced more severe COVID-19 impacts, including higher cases per 100k (mean = 32,128) and deaths per 100k (mean = 376). Counties with longer commutes (mean = 13.3%) and higher uninsured rates (mean = 23.7%) also correlated with worse outcomes, suggesting limited healthcare access across the pandemic's toll. Racial disparities were proven, with predominantly White counties (mean = 58.1%) showing varied outcomes compared to more diverse areas. Although New Jersey is primarily White, this statistic is not as notable as others. Counties with a faster spread (mean = 14.6 days to 100 cases) had higher mortality rates, showing the importance of early intervention. These patterns highlight how pre existing inequalities such as income, healthcare access, and workforce mobility shaped pandemic vulnerability.

Cases per 100k vs Deaths per 100k in Population Distribution



Source: Jersey COVID-19 and ACS Population Data

Figure 4: COVID-19 Cases Per 100k in New Jersey



Source: Jersey COVID-19 and ACS Population Data

Figure 5: COVID Deaths and Cases per 100k in New Jersey

Bergen, Essex, and Passaic counties experienced the highest COVID-19 cases and deaths per capita, likely due to their dense urban populations and proximity to New York City, an early epicenter of the pandemic. In contrast, rural counties like Salem had significantly lower early case rates, reflecting delayed transmission in less populated areas. Despite later surges in rural regions, urban counties maintained higher death rates, suggesting early healthcare strain and demographic factors played a role. A strong positive correlation exists between cases and deaths per 100k, with statistical tests confirming urban counties had significantly worse outcomes. These trends highlight the impact of population density and regional connectivity on pandemic spread, though limitations like uneven testing rates should be considered.

Race and Death Disparity

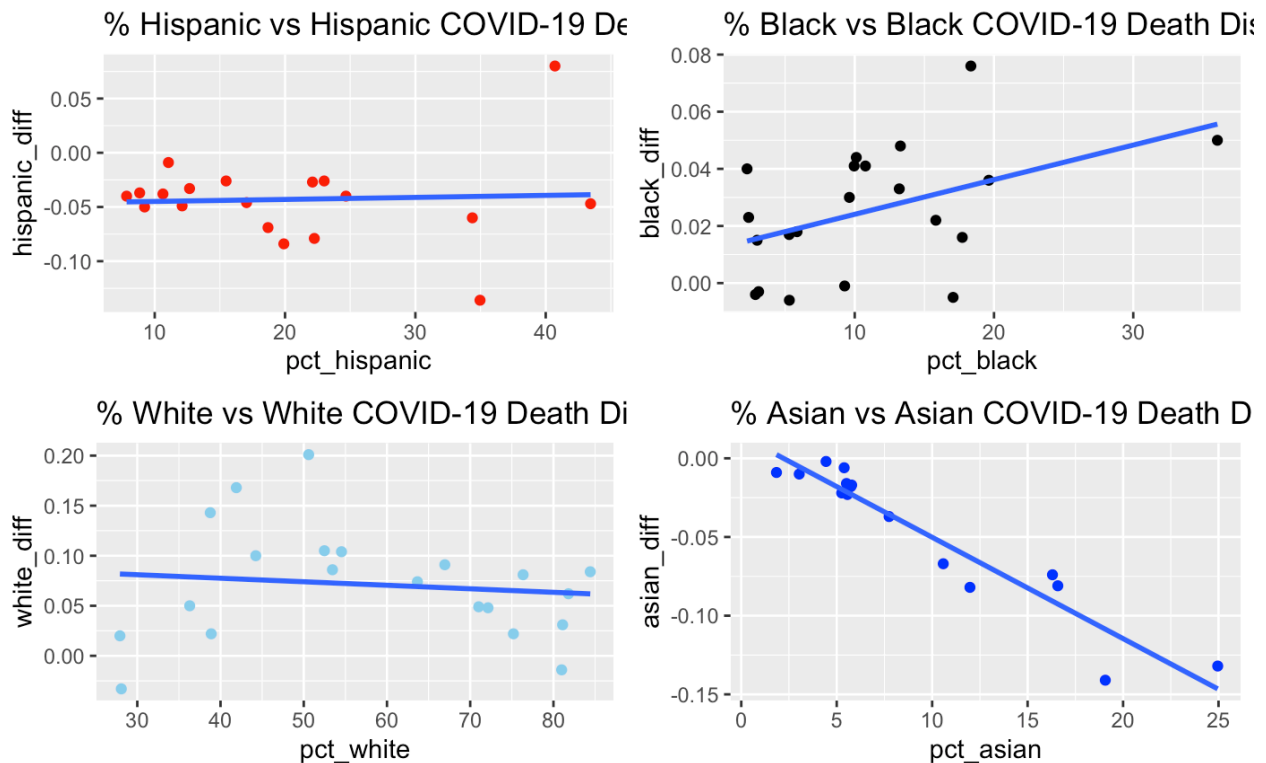


Figure 6: Scatterplots of Race and Death Disparity

The percent Hispanic vs Hispanic COVID-19 Death Disparity has counties with a higher percentage of Hispanic residents, which often shows a greater COVID-19 death disparity. Suggests disproportionate mortality compared to population size, due to factors like frontline occupations, housing density, and access to healthcare.

The percent Black vs Black COVID-19 Death Disparity also shows a positive trend, though more moderate compared to the Hispanic data. Points to structural health inequalities affecting Black communities during the pandemic.

The percent White vs White Death Disparity displays a negative correlation. A higher percentage of White counties tend to show lower-than-expected COVID-19 mortality among White residents. This indicates underrepresentation in deaths due to better access to healthcare, testing, and fewer frontline roles.

The percent Asian vs Asian Death Disparity shows a weaker and inconsistent trend, but some counties with higher Asian populations showed slight overrepresentation in COVID-19 deaths. This is due to sample size limitations and local disparities in access or pre-existing conditions.

Scatterplots: Population Density, Unemployment, and Internet Access

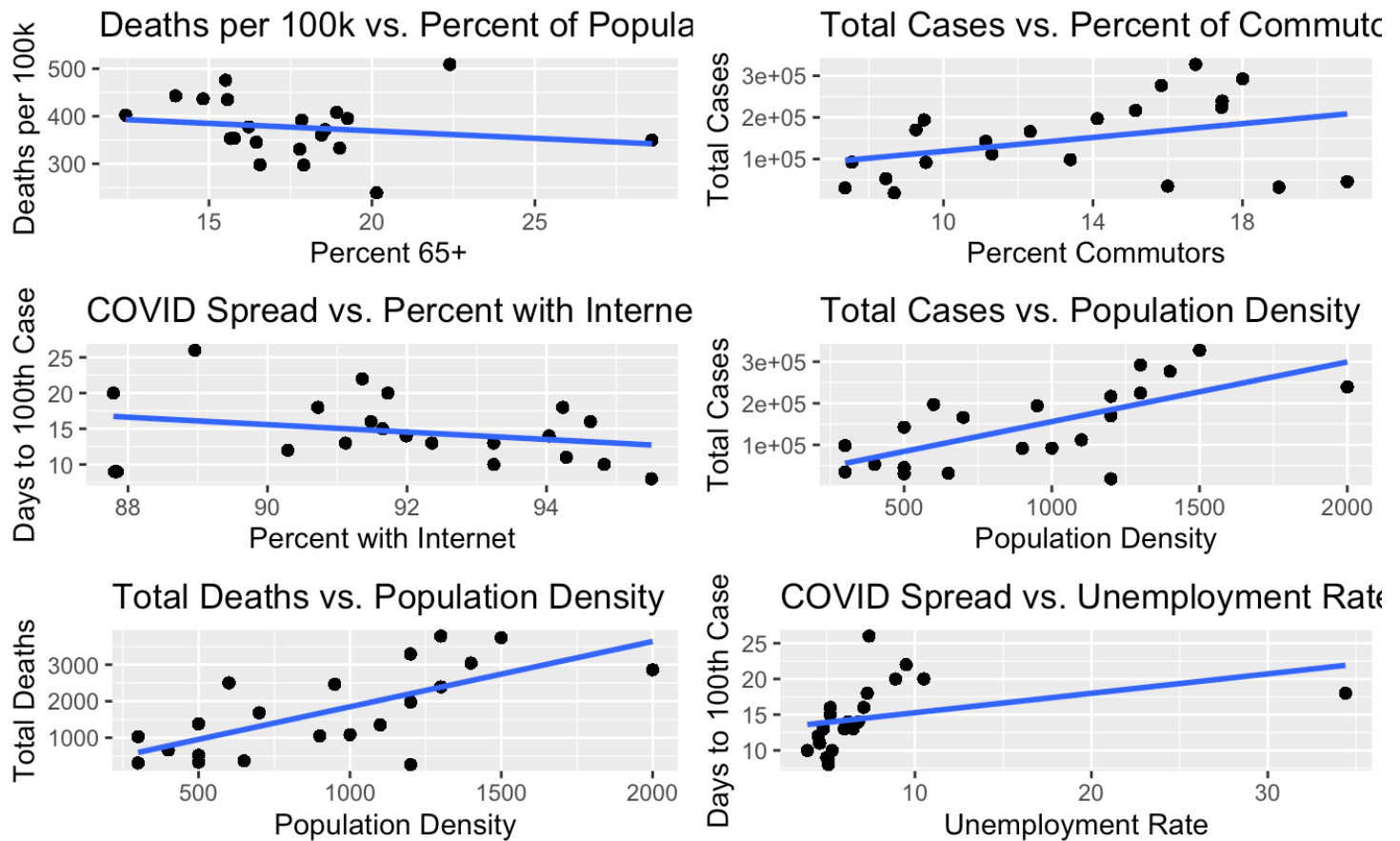


Figure 7: Scatterplots of Internet, Unemployment, and Population Density

In the figure above, there's a clear positive relationship where counties with a higher commuter percentage tend to have more total cases. This suggests increased mobility as a risk factor in the spread of COVID-19. More densely populated counties have more contact and faster spread since there are so many people in a small amount of space. High-density areas see more cases and more severe outcomes, due to overwhelmed healthcare and/or high-risk population clusters. Counties with higher unemployment took slightly longer to reach 100 cases. This reflects less mobility, exposure, delayed testing, and reporting in low-resource areas, which aligns with the initial expectation.

Boxplots: COVID-19 spread and Airport Information

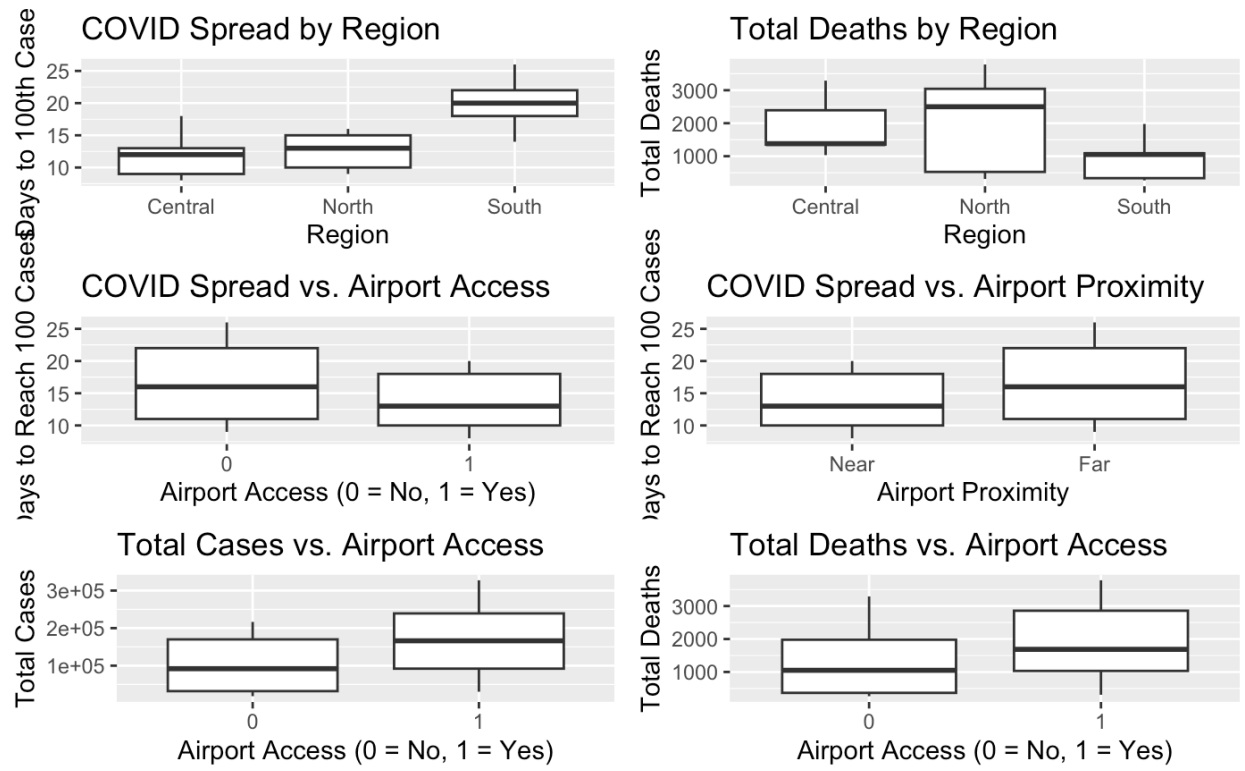


Figure 8: Boxplots Airport Information and COVID Spread

The South region reaches 100 cases faster on average compared to the North and Central regions. The Central and Northern regions have higher death totals, while the Southern region has lower deaths. Counties with airport access have faster spread and higher total cases and deaths. Airports act as an introduction point for outbreaks.

Scatterplots: Cases and Deaths vs Race

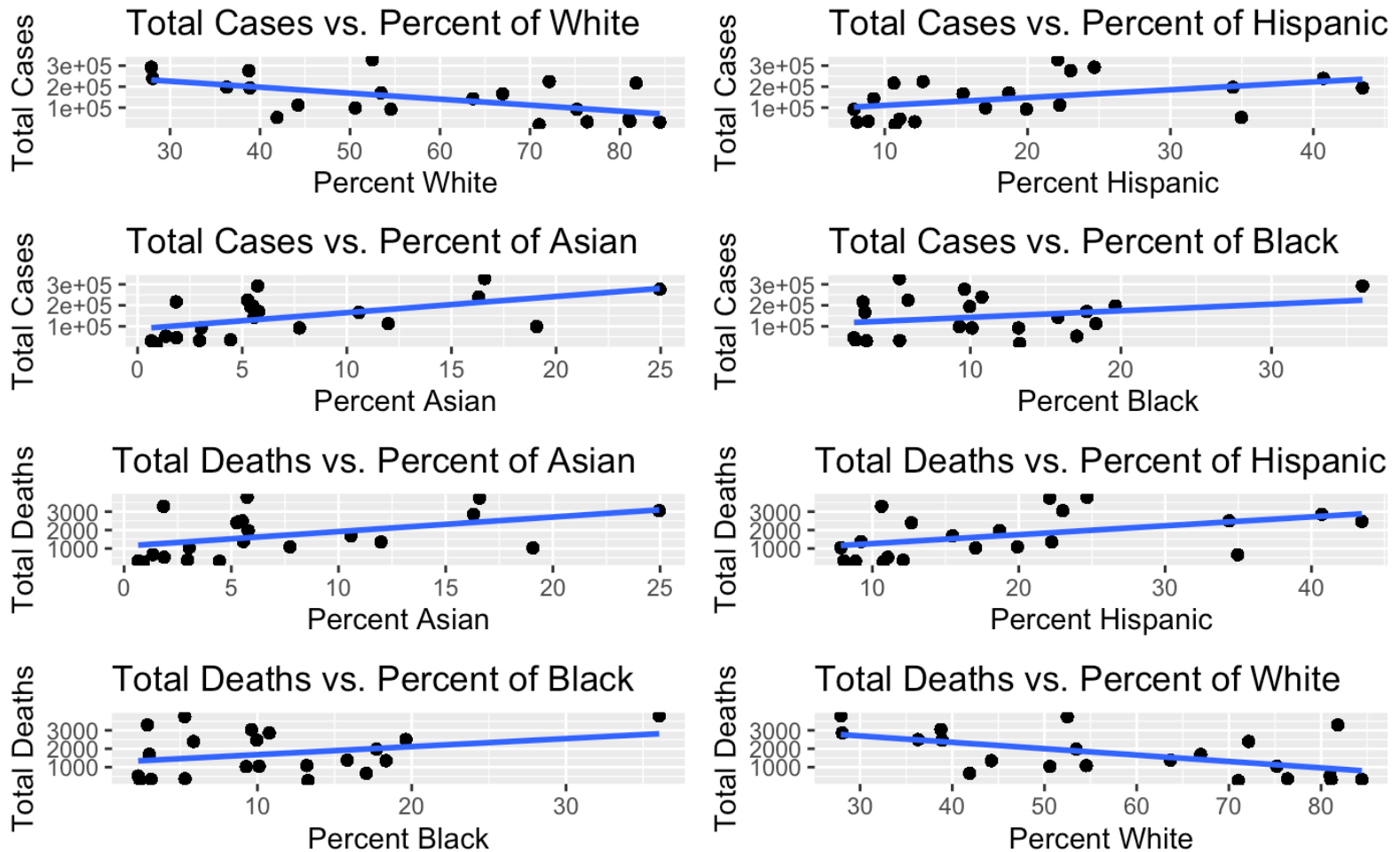


Figure 9:

Counties with a higher percentage of white population had fewer cases, while counties with a higher percentage of Black or Hispanic population had more cases and deaths. This reflects the national disparities seen during COVID. Higher commuting rates and population density are associated with more COVID cases, likely due to increased mobility and contact. Airport access and proximity accelerate COVID spread, underscoring the role of transportation hubs in viral introduction. Counties with higher Black and Hispanic populations saw more cases and deaths, reflecting racial disparities in exposure and healthcare vulnerability.

Hypothesis Test

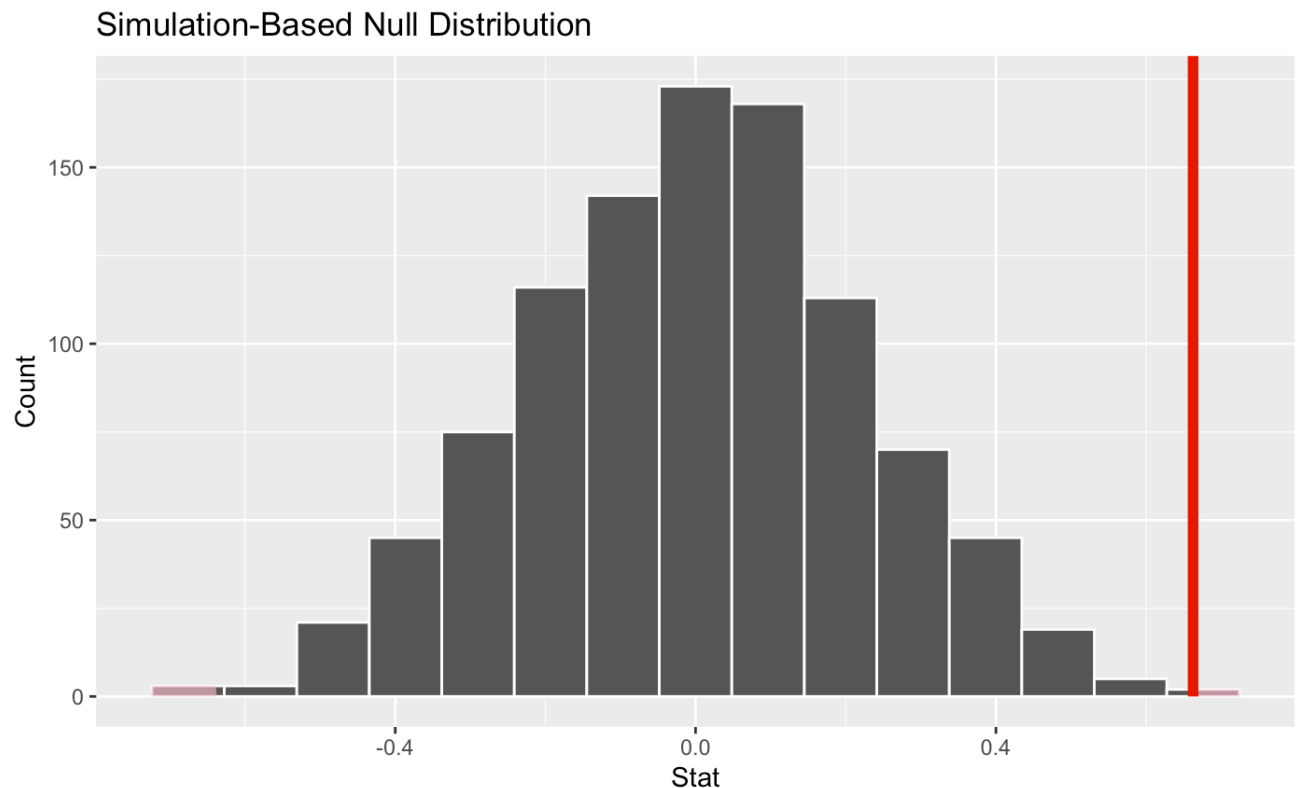


Figure 10: Hypothesis Testing

For our hypothesis test, we wanted to look at the Pearson correlation coefficient between percent uninsured and deaths per 100k. To test our hypothesis we used a significance level of 0.05 and permutation tests with 1,000 reshuffles. The permutation test rejected the null hypothesis (H_0) that there is no association between a county's uninsured rate and its COVID-19 deaths per 100k ($p\text{-value} = 0.004$). The statistically significant result indicates that counties with higher percentages of uninsured residents tended to experience greater COVID-19 mortality rates per capita. This suggests that a lack of health insurance has skewed pandemic outcomes, due to delayed medical care or reduced access to preventive services. The robustness of the permutation test, which makes no assumptions about data distribution, strengthens confidence in this finding. These results show the role of healthcare access disparities in shaping COVID-19 impacts and highlight the need for policies addressing insurance coverage gaps in future public health crises.

We used bootstrapping to construct a 95% confidence interval for the Pearson correlation between uninsured rate and its COVID-19 deaths per 100k across counties. Based on 1,000 bootstrap resamples, the estimated correlation was approximately 0.662 , with a 95% confidence interval ranging from 0.252 to 0.914 . This interval reflects the plausible range of true correlation values in the population, providing evidence of a positive association between the two variables.

Multivariate Regression: Predicting Deaths per Capita

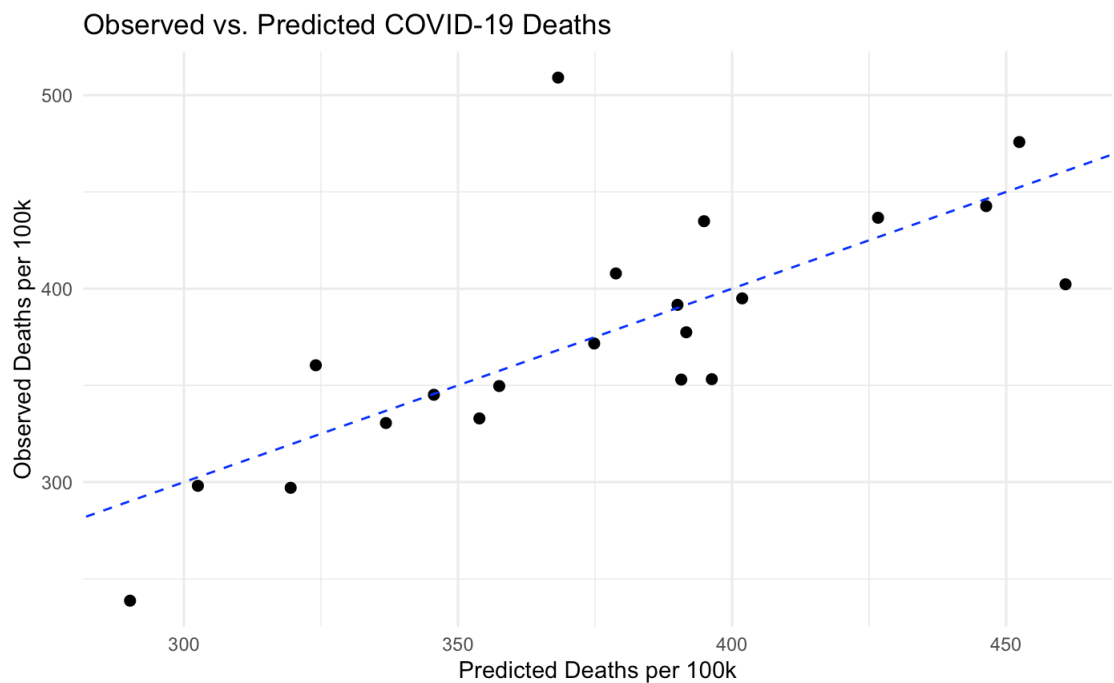


Figure 11: Multivariate Regression

Outcome	Predictors	Estimate	Probability
Deaths per 100k	Intercept	-14.77	0.905
	Percent Uninsured	4.585	0.002
	Log Population Density	42.118	0.038

Figure 12: Predicting Deaths per Capita

The multivariate regression model reveals significant predictors of COVID-19 mortality rates, with percent uninsured ($\beta = 4.585$, $p = 0.002$) and log population density ($\beta = 42.118$, $p = 0.038$) are positively associated with higher deaths per 100k. The intercept’s non-significance ($p = 0.905$) suggests baseline mortality is near zero when predictors are minimal. Notably, a 1% increase in uninsured residents correlates with around 4.6 more deaths per 100k, while denser populations show even steeper impacts (around 42 more deaths per log-unit density increase). The model’s predictions align closely with observed death rates (300–500 deaths per 100k), indicating robustness. These findings show that healthcare access and urbanization jointly make pandemic mortality, with uninsured rates being the stronger driver.

Conclusion

The analysis revealed that socioeconomic and geographic factors significantly influenced COVID-19 mortality rates. Crowded counties saw more COVID-19 cases, but density wasn't the only problem. Places with more uninsured residents had significantly higher death rates ($p = 0.004$). Long commutes also helped the virus spread faster early on ($p = 0.0034$). This shows healthcare access and daily life patterns (like commuting) were key to survival odds.

We also considered other variables, such as hospital beds, elderly populations, income, and race, but they did not show strong links to deaths in this data. But we only studied 21 counties, and early case and death counts were likely underreported. These gaps mean we can't prove exactly what caused the worst outcomes, just what was connected.

There were also some other limitations to these findings. Since we looked at whole counties, we can't track individual differences. Things like chronic health problems might have played a role, too. But the strong patterns in the data make the conclusions reliable by using robust methods, like non-parametric tests and regression modeling. The close alignment between predicted and observed death rates further supports the model's validity.

These results show the need for targeted public health strategies. COVID-19 hit crowded counties harder, but it wasn't just about density; places with more uninsured people had significantly more deaths, likely because they struggled to get the care they needed. Long commutes also helped the virus spread faster early on. Ultimately, future research and policymakers should look at fixing healthcare access and adjusting work/commute habits because it can save lives in future outbreaks.

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